Quantification: 682_z29
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 14 |
| interior lacuna count | 12 |
| frame-edge lacuna count | 2 |

Interior lacunae only: mean, median and sample SD (n = 12)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1650.75 | 1511.5 | 948.377 | px^2 |
| major_axis_length_px | lacuna shape | 74.0009 | 74.236 | 22.3194 | px |
| minor_axis_length_px | lacuna shape | 27.9394 | 29.9101 | 9.5919 | px |
| aspect_ratio | lacuna shape | 2.805 | 2.6218 | 0.775 | unitless |
| eccentricity | lacuna shape | 0.9144 | 0.9241 | 0.0615 | unitless |
| solidity | lacuna shape | 0.8593 | 0.8693 | 0.0551 | unitless |
| roots_count | headline | 8.3333 | 8.5 | 3.7009 | count |
| ring_length_r30_px | headline | 321.75 | 337 | 124.77 | px |
| ring_length_r60_px | reported | 799.083 | 849.5 | 313.994 | px |
| owned_length_px | ownership-dependent | 2442.49 | 2352.35 | 1917.49 | px |
| edge_count | ownership-dependent | 107.083 | 94.5 | 89.4046 | count |
| mean_edge_length_px | ownership-dependent | 23.8341 | 22.9862 | 3.5854 | px |
| perimeter_px | normalised | 197.604 | 208.43 | 59.2561 | px |
| ring_area_r30_px2 | normalised | 7853.83 | 8153 | 2003.08 | px^2 |
| ring_area_r60_px2 | normalised | 19782.8 | 19785 | 4716.97 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9948 | 1 | 0.018 | unitless |
| in_frame_fraction_r60 | normalised | 0.9826 | 1 | 0.0559 | unitless |
| ring_density_r30 | normalised | 0.0406374 | 0.0402829 | 0.0108351 | px^-1 |
| ring_density_r60 | normalised | 0.0397348 | 0.0395364 | 0.00950225 | px^-1 |
| roots_per_100px_perimeter | normalised | 4.2347 | 4.0412 | 1.4256 | per 100 px |
| ring_attached_length_r30_px | network v2 | 291.167 | 298.5 | 123.874 | px |
| ring_attached_length_r60_px | network v2 | 663.417 | 641.5 | 330.81 | px |
| ring_length_w_r30_px | network v2 | 360.743 | 373.108 | 137.027 | px |
| ring_length_w_r60_px | network v2 | 898.556 | 942.947 | 352.262 | px |
| sholl_crossings_r10 | network v2 | 7.9167 | 7.5 | 2.7455 | count |
| sholl_crossings_r20 | network v2 | 10.0833 | 11 | 3.8248 | count |
| sholl_crossings_r30 | network v2 | 11.3333 | 10.5 | 3.9619 | count |
| loop_count | loops, report only | 11.25 | 5.5 | 14.2135 | count |
| loop_edge_share | loops, report only | 0.2422 | 0.238 | 0.2152 | share |
| ring_width_mean_r30_px | width | 7.1961 | 7.1588 | 0.5658 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.0229e+06 | px^2 |
| total_skeleton_length_px | 37631 | px |
| canalicular_length_density_per_px | 0.0367884 | px^-1 |
| junction_count | 822 | count |
| junction_density_per_px2 | 0.0008036 | px^-2 |
| lacuna_count | 14 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.369e-05 | px^-2 |
| skeleton_component_count | 236 | count |
| mean_component_length_px | 159.453 | px |
| median_component_length_px | 21 | px |
| field_length_density_w_per_px | 0.0415437 | px^-1 |
| field_density_without_flagged_per_px | 0.0370778 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 513 | px |
| unattached_component_count | 32 | count |
| canalicular_length_density_attached_per_px | 0.0367884 | px^-1 |
| field_length_density_with_unattached_per_px | 0.03729 | px^-1 |
| loop_count | 165 | count |
| loop_edge_share | 0.3414 | share |
| edge_count | 1822 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.6927 | px |
| width_median_px | 8 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 35121 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2083 | 86.508 | 34.896 | 2.479 | 0.915 | 0.8266 | 1.5435 | 13.4983 | 99.4023 | True |
| 2 | 2213 | 89.1614 | 32.1486 | 2.7734 | 0.9327 | 0.8778 | -1.1291 | 134.914 | 921.292 | False |
| 3 | 446 | 49.6136 | 12.2658 | 4.0449 | 0.969 | 0.7866 | -1.4445 | 249.987 | 449.359 | False |
| 4 | 3140 | 120.617 | 34.4194 | 3.5043 | 0.9584 | 0.8885 | 1.5609 | 431.048 | 220.093 | False |
| 5 | 2837 | 88.9183 | 41.496 | 2.1428 | 0.8844 | 0.9229 | -1.5453 | 532.043 | 493.055 | False |
| 6 | 1634 | 75.9278 | 30.0551 | 2.5263 | 0.9183 | 0.8211 | -1.4797 | 555.483 | 357.187 | False |
| 7 | 3781 | 121.501 | 43.6088 | 2.7862 | 0.9334 | 0.8771 | 1.4272 | 590.482 | 55.0987 | True |
| 8 | 1399 | 81.52 | 23.1918 | 3.515 | 0.9587 | 0.8541 | 1.5542 | 716.135 | 181.57 | False |
| 9 | 663 | 61.7115 | 15.8655 | 3.8897 | 0.9664 | 0.7408 | -1.1603 | 728.596 | 861.06 | False |
| 10 | 513 | 41.8409 | 17.3844 | 2.4068 | 0.9096 | 0.8607 | 1.3405 | 830.565 | 75.5867 | False |
| 11 | 1130 | 64.3164 | 23.6688 | 2.7173 | 0.9298 | 0.8835 | 1.4062 | 842.292 | 125.237 | False |
| 12 | 1244 | 49.7173 | 33.3792 | 1.4895 | 0.7411 | 0.9325 | -1.542 | 904.6 | 760.807 | False |
| 13 | 2966 | 92.1224 | 41.6334 | 2.2127 | 0.8921 | 0.895 | -1.4899 | 920.401 | 307.385 | False |
| 14 | 1624 | 72.5441 | 29.7651 | 2.4372 | 0.9119 | 0.8476 | -0.945 | 978.497 | 437.615 | False |

Network per lacuna

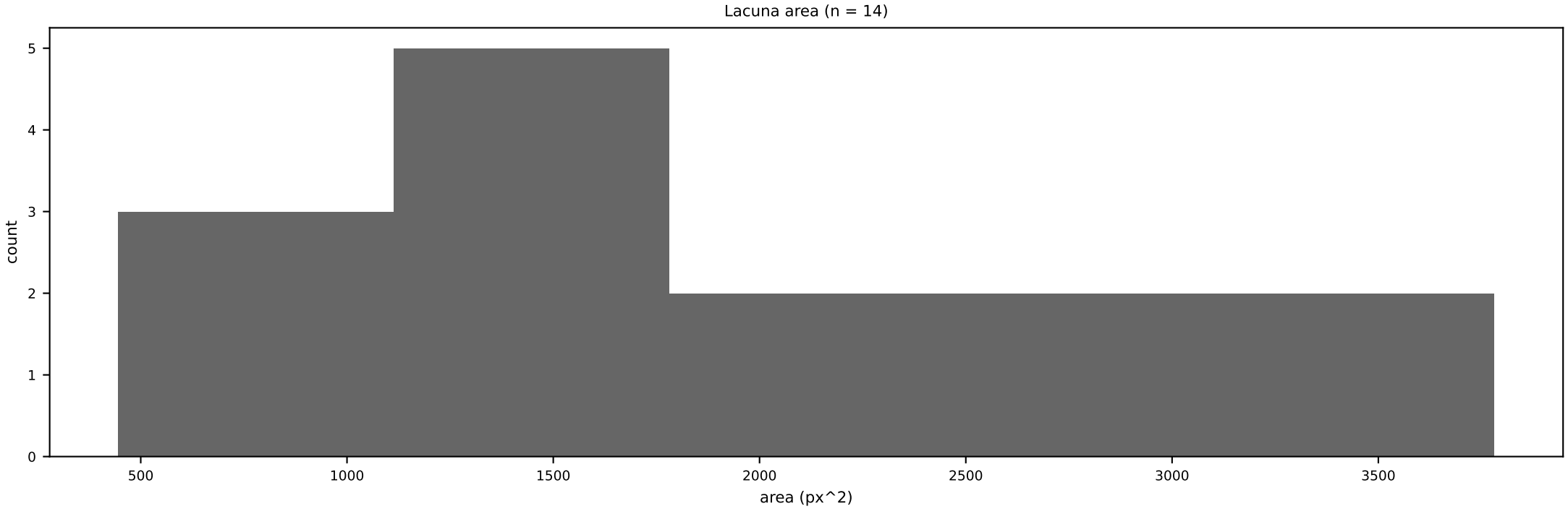
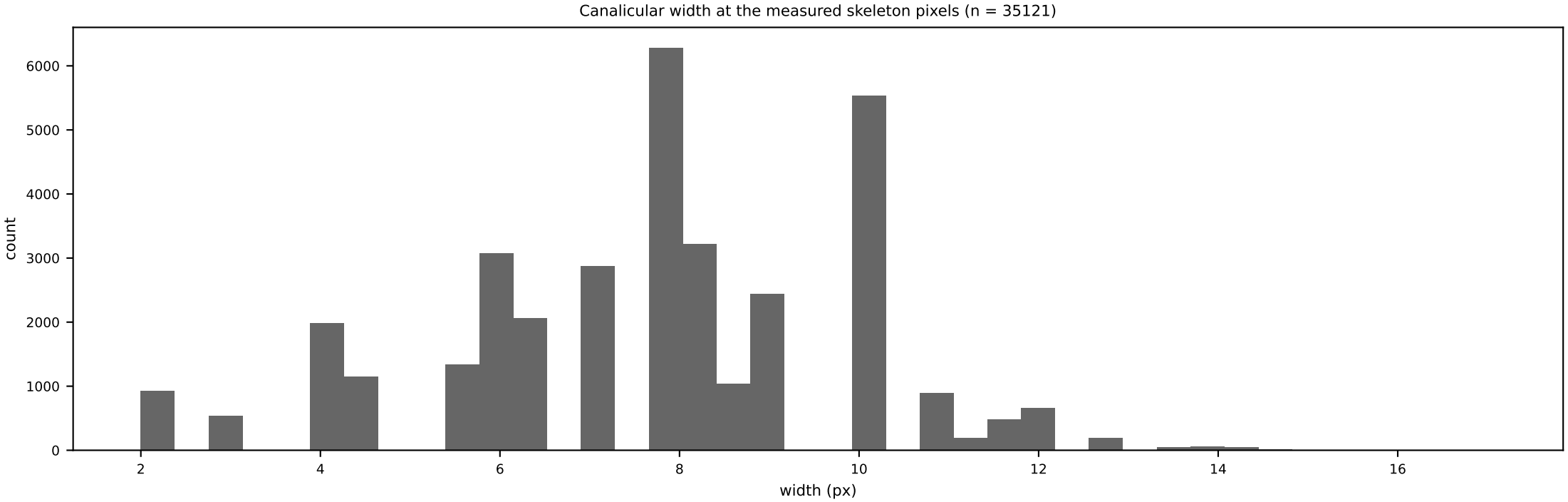
| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 4 | 192 | 359 | 1495.72 | 79 | 18.9332 | |
| 2 | 17 | 536 | 1315 | 6300.68 | 276 | 22.8286 | 7.2892 |
| 3 | 6 | 361 | 940 | 4633.43 | 213 | 21.7532 | 7.3901 |
| 4 | 12 | 476 | 1334 | 4428.32 | 228 | 19.4225 | 7.7077 |
| 5 | 7 | 313 | 842 | 2980.13 | 118 | 25.2553 | 7.1922 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 9 | 432 | 911 | 2707.02 | 120 | 22.5585 | 7.3207 |
| 7 | 11 | 392 | 836 | 1426.67 | 64 | 22.2918 | 7.4798 |
| 8 | 8 | 376 | 895 | 1997.69 | 73 | 27.3657 | 6.2524 |
| 9 | 9 | 214 | 646 | 2834.25 | 116 | 24.4332 | 6.9231 |
| 10 | 5 | 174 | 466 | 509.161 | 22 | 23.1437 | 6.8561 |
| 11 | 5 | 190 | 476 | 507.446 | 16 | 31.7154 | 8.5704 |
| 12 | 3 | 176 | 516 | 645.47 | 34 | 18.9844 | 6.8366 |
| 13 | 9 | 385 | 857 | 1397.99 | 52 | 26.8844 | 6.8891 |
| 14 | 10 | 228 | 391 | 368.291 | 17 | 21.6642 | 7.1254 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 262.096 | 5607 | 13819 | 0.5809 | 0.5584 | 0.0342429 | 0.0259787 | 1.5262 |
| 2 | 229.522 | 9151 | 23871 | 1 | 0.999 | 0.0585728 | 0.0550878 | 7.4067 |
| 3 | 120.811 | 6134 | 17911 | 1 | 1 | 0.0588523 | 0.0524817 | 4.9664 |
| 4 | 309.279 | 11331 | 28206 | 1 | 1 | 0.0420087 | 0.0472949 | 3.88 |
| 5 | 235.238 | 9387 | 21888 | 1 | 1 | 0.033344 | 0.0384686 | 2.9757 |
| 6 | 214.167 | 8637 | 20090 | 1 | 1 | 0.0500174 | 0.0453459 | 4.2023 |
| 7 | 322.836 | 9940 | 22198 | 0.8402 | 0.7724 | 0.0394366 | 0.0376611 | 3.4073 |
| 8 | 229.48 | 8595 | 22042 | 1 | 1 | 0.0437464 | 0.0406043 | 3.4861 |
| 9 | 156.853 | 6922 | 19429 | 1 | 1 | 0.0309159 | 0.0332493 | 5.7379 |
| 10 | 109.498 | 4245 | 11066 | 1 | 0.9865 | 0.0409894 | 0.042111 | 4.5663 |
| 11 | 161.817 | 5491 | 13376 | 1 | 1 | 0.0346021 | 0.0355861 | 3.0899 |
| 12 | 147.196 | 6914 | 19480 | 1 | 1 | 0.0254556 | 0.0264887 | 2.0381 |
| 13 | 254.693 | 9728 | 23664 | 1 | 1 | 0.0395765 | 0.0362154 | 3.5337 |
| 14 | 202.693 | 7711 | 16371 | 0.9376 | 0.8055 | 0.0295682 | 0.0238837 | 4.9336 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 85 | 150 | 219.51 | 395.664 | 2 | 6 | 6 |
| 2 | 528 | 1263 | 590.333 | 1486.51 | 12 | 17 | 18 |
| 3 | 319 | 940 | 407.806 | 1046.07 | 8 | 11 | 11 |
| 4 | 450 | 1208 | 539.304 | 1503.74 | 13 | 14 | 16 |
| 5 | 278 | 702 | 338.409 | 926.056 | 7 | 12 | 13 |
| 6 | 361 | 719 | 481.534 | 1022.01 | 9 | 11 | 15 |
| 7 | 379 | 771 | 440.563 | 939.968 | 10 | 12 | 12 |
| 8 | 358 | 644 | 417.664 | 999.11 | 10 | 12 | 10 |
| 9 | 191 | 482 | 252.037 | 750.382 | 5 | 7 | 9 |
| 10 | 161 | 377 | 194.51 | 519.99 | 6 | 5 | 6 |
| 11 | 190 | 422 | 220.238 | 534.647 | 5 | 7 | 8 |
| 12 | 127 | 316 | 199.995 | 597.085 | 5 | 7 | 8 |
| 13 | 342 | 639 | 432.363 | 959.838 | 9 | 13 | 15 |
| 14 | 189 | 249 | 254.723 | 437.233 | 6 | 5 | 7 |

682_z29: distributions, pre-validation, px